

Molecular models for recombination and disproportionation of radicals¹

SIDNEY W. BENSON

Hydrocarbon Research Institute, University of Southern California, Los Angeles, CA 90007, U.S.A.

Received July 5, 1982

This paper is dedicated to Professor Harry E. Gunning on the occasion of his 65th birthday

SIDNEY W. BENSON. Can. J. Chem. **61**, 881 (1983).

The modified Gorin model of recombination as a long range interaction of weakly bound radicals, constrained only by steric forces, is rederived to give a collision impact parameter 11% larger than $r_{\max}$, the separation at the centrifugal maximum. This larger impact parameter resolves some of the earlier difficulties with secondary and tertiary radicals. By imposing a new constraint, that the orientation of the radicals be such as to permit exchange forces to act, a new model is developed in which $k_r = 1/4 Z_r \alpha$, where α represents the product of fractions of active surface areas of each radical available for bonding. Excellent agreement is obtained in estimating values of k_r .

Analysis of disproportionation is made to show that it cannot be described by the tight transition state but must also be governed by similar, long range exchange forces between the bond being attacked and the same "active surface" of the abstracting radical. Using an ad hoc radius for $C \cdots H-C$ equal to 4 Å it is possible to reproduce the data on alkyl radical disproportionation to within $\pm 10\%$.

The model predicts a small decrease in rate of recombination with increasing temperature. This decrease is more pronounced for bulkier radicals and at higher temperatures (>900 K).

SIDNEY W. BENSON. Can. J. Chem. **61**, 881 (1983).

On a procédé à une nouvelle dérivation du modèle de recombinaison de Gorin modifié en tant qu'interaction à longue distance des radicaux faiblement liés et encombrés uniquement par des forces stériques dans le but de donner un paramètre d'impact collisionnel de 11% supérieur au $r_{\max}$, la séparation au maximum centrifuge. Ce plus grand paramètre d'impact résout quelques-unes des difficultés antérieures rencontrées avec les radicaux secondaires et tertiaires. En imposant une nouvelle contrainte, à savoir que l'orientation des radicaux doit être telle qu'elle permet aux forces d'échanges d'agir, on a mis au point un nouveau modèle dans lequel $k_r = 1/4 Z_r \alpha$, où α représente le produit des fractions des surfaces actives de chaque radical disponible pour la liaison. On obtient un excellent accord en évaluant k_r .

On a analysé la dismutation pour montrer qu'on ne peut pas la décrire par l'état de transition intime mais qu'elle doit aussi être gouvernée par des forces d'échange à longue distance semblables entre le lien attaqué et la même "surface active" du radical qui provoque l'élimination. En utilisant un rayon "ad hoc" pour $C \cdots H-C$ égal à 4 Å, il est possible de reproduire les données relatives à la dismutation des radicaux alkyles avec une marge d'erreur de $\pm 10\%$.

Le modèle prédit une légère diminution de la vitesse de recombinaison quand la température augmente. Cette diminution est plus prononcée pour les radicaux encombrés et se manifeste à des températures plus élevées (>900 K).

[Traduit par le journal]

Introduction

Transition State Theory permits us to model chemical reactions as though we were modeling equilibria. That is, we can treat the metastable short-lived transition state complex as though it were a normal molecule. Almost two decades of testing such models against experimental data have seen the development of what are known as "tight" and "loose" transition states. The loose transition state was originally developed by Gorin (1) to describe the recombination of methyl radicals and has since been generalized to describe all radical-radical recombinations (2). The loose transition state resembles two weakly interacting radicals, with their active centers separated by $r^\ddagger$ about 2.6 to 3.2 times the normal bonding distance r_0 . The relation employed has been derived (2) as:

$$[1] \quad r^\ddagger/r_0 = (6V_0/RT)^{1/6}$$

where V_0 is the potential energy of the bond being formed and T is the absolute temperature. The distances so obtained for $r^\ddagger$ correspond, curiously enough, to van der Waals' distances for many molecules. At $r^\ddagger$ the recombining centers have a potential of interaction of about $RT/2$, a very weak interaction.

At the other end of the reaction spectrum is the tight transition state which has been found to characterize well the transition states² for metathesis reactions between molecules and radicals, at least for those reactions whose intrinsic activation energies exceed 1–2 kcal. With metathesis we include the addition of radicals or atoms to unsaturated systems which may be looked upon as the metathesis of π -bonds as opposed to σ -bonds which characterize atom transfers. From an electron point of view both metathesis and addition reactions are 3-center, 3-electron systems:



The bonding distances in these reactions are taken as 0.3 Å larger than normal bond lengths and stretching and bending force constants are assumed to be 1/2 their normal values so that the corresponding frequencies are about 0.7 of their normal value.

¹This work has been supported by grants from the National Science Foundation (CHE 7826623) and the U.S. Army Research Office (DAAG 27-76-G-0195).

²In general it is possible to predict the A-factors and their temperature dependence within an uncertainty of a factor of 2 or less. The chief uncertainty arises from the uncertainty in knowing whether one is dealing with a linear or bent transition state $A \cdots B \cdots C$.

In the present paper we will discuss a model for radical-radical interactions which will have characteristics of both the loose and tight transition states depending on whether we are considering recombination or disproportionation, respectively. What is most striking about these reactions is that they appear to have zero activation energy for both channels, recombination and disproportionation. Highly branched radicals such as *tert*-butyl appear to deviate from this general rule, showing a small decrease in rate of recombination with increasing temperature (6). Also striking about radical-radical reactions is that the rate constants generally fall between the limits of $10^{10.5}$ to $10^{8.0}$ L/mol-s, for all radical species. In fact, a mean value of $10^{9.5 \pm 0.5}$ would fit 90% of all systems studied.

Recombination of radicals; collision dynamics

Despite intensive efforts extending over a 30-year period, few if any radical-radical rate constants are known to better than a factor of 2 in accuracy. While recent studies show a greater reproducibility, they still show differences of this order between different experimental techniques (3). The most extensively studied system has been methyl recombination, for which 11 separate studies since 1967 yield values for k_r centered about $10^{10.4}$ L/mol-s near 300 K. Values range from $10^{10.38}$ to $10^{10.76}$ (3).³ Studies at higher temperatures suggest a small decrease in k_r with increasing temperatures. This is difficult to separate from the fall-off in rate with pressure at temperatures above 900 K.

Because of the fact that k_r values are near collision frequencies, it has been useful to compare k_r with collision frequencies corrected for electronic degeneracies. If we write k_r in these terms:

$$[2] \quad k_r = g_e Z_r \alpha$$

then g_e is the ratio of electronic partition functions $g^\ddagger / g_1 g_2$ for the transition state and radicals. It is usually equal to 1/4, corresponding to the spin factors for radicals which has spin 1/2 forming a singlet recombination product. Z_r is the collision frequency using $r^\ddagger$ (eq. [1]) for the collision diameter, while α is the steric factor for the collision and reflects the fact that, at the given separation of active centers $r^\ddagger$, not all angular orientations of both radicals are possible. In simple cases it is possible to calculate α directly (2) as a product of the fractions of the solid angles available to each species in the transition state. This modified Gorin model has been called "the restricted, free rotor model" and has been derived from an oversimple model of two point particles moving under the influence of central forces (2). The equation of motion for such a system can be described by:

$$[3] \quad E_0 = \frac{1}{2} \mu (\dot{r})^2 + E_0 \frac{b^2}{r^2} - 2 \frac{V_0 r_0^6}{r^6} = \frac{1}{2} \mu (\dot{r}_0)^2$$

where r is the radial separation of the point particles with initial relative kinetic energy E_0 , impact parameter b , normal bond length r_0 , and potential energy of dissociation V_0 . $E_0 b^2 / r^2$ represents the centrifugal force in the polar trajectory; μ is the reduced mass of the system. The term in V_0 assumes that the valence forces can be represented by an r^{-6} potential at large distances.

In the usual derivation one calculates the forces by differentiation of the potential energy and calculates $r_{\max}$, the separation

at which the net force is zero. This is the centrifugal maximum in the potential function where the centrifugal and bonding forces are just equal. At smaller distances the bonding forces are larger, while at longer distances the repulsive centrifugal forces are dominant.

The trajectory described by eq. [3] (which neglects shorter range repulsive forces) is a complex, but smooth, path characterized for each initial set of values of E_0 and b by a distance of closest approach r_c . If, in the course of the trajectory, $r \leq r_{\max}$ and $\dot{r}_{\max} \leq 0$, then, since the forces are attractive in this region, $\dot{r}$ will remain negative and the trajectory will speed in towards the origin with increasing radical velocity. Note that, since we start with separated particles, $\dot{r}_0$ is initially negative. These trajectories resemble very much those for the Langevin Model for ion-molecule encounters which, in fact, historically provided the inspiration for the model (1). For real molecules the inclusion of short range repulsive forces in eq. [3] will prevent encounters at the origin.

While $r_{\max}$ is the minimum collision diameter required for a close encounter it is not the impact parameter for collision. One can calculate the latter by imposing two conditions on the collision. We want to know, for a given E_0 , what impact parameter $b^\ddagger$ will satisfy the condition that $r_{\max} \leq r_c$. On applying this to eq. [3], we find a result very similar to eq. [1]:

$$[4] \quad b^\ddagger = 1.5(V_0/E_0)^{1/6}$$

If we take $\langle E_0 \rangle = RT$, then

$$[5] \quad b^\ddagger = 1.5(V_0/RT)^{1/6}$$

Although functionally the same as eq. [1], the coefficient is about 11% larger and hence the cross section is about 23% larger.

While functionally of the correct form, eq. [5] does not represent the result of proper averaging of $b^\ddagger(E)$ over a Boltzmann population. This latter is given by:

$$[6] \quad \langle b^\ddagger(E) \rangle = \int_0^\infty b^\ddagger P(E) dE$$

and using the appropriate $P(E)$ (9)⁴ we find:

$$[7] \quad \langle b^\ddagger(E) \rangle = 1.52 \left(\frac{V_0}{RT} \right)^{1/6}$$

a result which differs from eq. [5] by about 1.5%

A somewhat larger correction is found when we take the proper average of Z_r over a Boltzmann population. Z_r is given by (9):

$$[8] \quad Z_r = \pi (b^\ddagger)^2 c_r$$

where c_r is the relative translational velocity of the colliding pair ($\dot{r}_0$ in eq. [2]). Replacing $b^\ddagger$ by its value from eq. [4] and setting $E_0 = 1/2 \mu c_r^2$, we have:

$$[9] \quad Z_r = \pi (1.5)^2 \left(\frac{2V_0}{\mu} \right)^{1/3} \langle c_r \rangle^{1/3}$$

The Boltzmann average is then:

$$[10] \quad \langle Z_r \rangle = 2.25 \left(\frac{2V_0}{\mu} \right)^{1/3} \langle c_r \rangle^{1/3}$$

⁴ It can be shown that

$$\langle C^n \rangle = \left(\frac{2RT}{\mu} \right)^{n/2} \frac{\Gamma[(n+3)/2]}{\Gamma(3/2)}$$

³ All the literature to the end of 1977 will be found in Vol. II; Vol. I summarizes data on radical metathesis.

$$[11] \quad = \frac{2.25}{(\mu/2)^{1/2}} (V_0)^{1/3} (RT)^{1/6} (1.02)$$

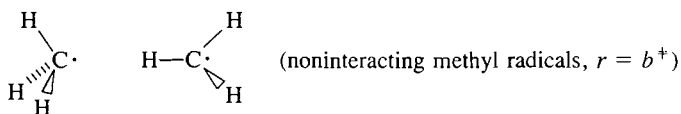
This result is 10% smaller than the result obtained by averaging c_r and $b^\ddagger$ separately and then substituting in eq. [8].

An observation on the dynamics of the encounter described by eq. [3] is appropriate. In the absence of any forces, $r_c = b^\ddagger$ is realized at $\theta = 90^\circ$ when the particles pass each other. During this entire trajectory $r \geq b^\ddagger$. In the presence of attractive forces, r will become less than b at $\theta \doteq 90^\circ$. We observe that $b^\ddagger = 1.11 r^\ddagger$ marks the boundary line between close and distant encounters.

Methyl radical recombination

For CH_3 recombination, $\Delta H_r^\circ = -90.4$ kcal/mol at 300 K (4). Correcting this to 0 K and for zero point energy we find $V_0 = 96$ kcal so that $r^\ddagger$ (300 K) = 4.84 Å and $\langle b^\ddagger \rangle = 5.37$ Å. Using $g_c = 1/4$ we derive $Z_r = 2.8 \times 10^{11}$ L/mol-s and $k_r = 7.0 \times 10^{10}$. Comparing this with the experimental mean value of 2.4×10^{10} at 300 K we see that $\alpha = 0.34$, implying that 0.66 of all the possible geometric conformations of two CH_3 groups with carbon atoms 5.37 Å apart will not lead to recombination. At this separation, however, we can find that the hindered free rotor model gives no angular correction, that is, $\alpha = 1$. Using a 1.2 Å van der Waals' radius for H, we find that the methyl radicals can rotate freely at $b^\ddagger$ (C—C) = 5.37 Å. We could conclude that the potential function is incorrect and that the true $b^\ddagger$ (C—C) = 3.42 Å which would then give the observed k_r for $\alpha = 1$. This, however, is not an acceptable solution. At 3.42 Å, the methyl radicals are appreciably constrained in their orientation and α would have to be appreciably less than 1.

We should like to suggest an approximate solution to this dilemma for which again a relatively simple model may be adopted. The hindered free-rotor model is a necessary but not a sufficient condition for radical recombination. In order for recombination to occur, exchange forces must be operative between the two radicals. This in turn requires an overlap between the unoccupied electron orbitals. A conformation of two CH_3 radicals in which a H-atom lies along the line of centers, while not involving any van der Waals' repulsion at $b^\ddagger$, $b^\ddagger$ (C—C) = 5.37 Å, also does not permit orbital overlap



and thus cannot lead to the weak attraction required by the Gorin Model. Such an analysis suggests a modified model in which we draw a sphere centered on the free radical atom; for $\dot{\text{C}}\text{H}_3$ it is $\dot{\text{C}}$. The radius of this sphere is taken as $r = b^\ddagger/2$. We can now assign the surface of this sphere as active or inactive, depending on whether it is occupied by free electron density (active) or closed shell electrons (inactive). A critical collision for recombination is then considered to occur when two such spheres make contact with their active surfaces touching. In such configurations orbital overlap may occur. When these active areas are in contact, attraction is possible and recombination can take place.

If we represent by β the fraction of the "surface area" of the CH_3 radical of radius = 5.37/2 = 2.68 Å (centered at C) which is active, then $(1 - \beta)$ is the inactive fraction and $\alpha \leq \beta^2$. The inequality is used since, for radicals more complex than CH_3 ,

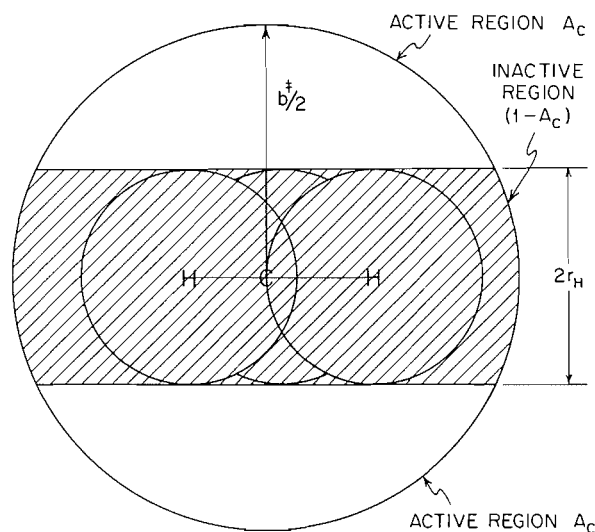


FIG. 1. Methyl radical in projection showing regions active and inactive (shaded) in recombination. r_H is the van der Waals' radius of the H atom; $b^\ddagger$ is the impact parameter for recombination. The third H atoms is behind the two shown.

nonbonded interactions between other portions of the radicals may inhibit contact between the active areas.⁵ Using an experimental $\alpha = 0.34$ for CH_3 recombination, we can estimate the active area as $\beta = 0.66$ and the excluded region due to the H atoms $(1 - A) = 0.34$ of the surface area. Figure 1 shows such a sphere for CH_3 .

Is this physically reasonable? A simple model suggests that it is. If we take, as a model of the CH_3 radical, one which is planar so that the H-atoms form a belt around the center then we can project the area of this belt as a zone, symmetrical about the center of the sphere with a width equal to the van der Waals' radius of H of 1.2 Å. This belt subtends a zenith angle of approximately 27° about either side of the equator of this 2.68 Å sphere centered on C, and the zone would subtend 0.45 of the total surface area for $\beta = 0.55$ and $\alpha = 0.30$, in excellent agreement. If we assign to CH_3 a tetrahedral geometry and make the same calculation but assume we have two active areas, a large one with azimuth angle of 83° and a smaller one on the back of the CH_3 with azimuth angle of 44° , the total active area is 58% and $\alpha = 0.34$, not much different. Both models assume that front and back of the CH_3 radical are available for overlap and yield a recombination value differing by about 20% from the "best" observed value.

Baulch and Duxbury (5) have recently made a survey and analysis of k_r for CH_3 radicals and proposed a preferred value of $\log k_r = 10.38$ in the range 250–420 K.

From an electron orbital point of view such a model corresponds to an orbital centered on carbon and occupied by a single electron. Pauli exclusion guarantees that this odd electron will be restricted to a volume in space outside the van der Waals' region occupied by the other bound electrons in the system, namely the electron-pairs in the adjacent C—H bonds. This would account for the "shadowing" effect of the neighboring bonds and the restriction of the "active" area or volume to that region in space available to the odd electron.

In Table I we list values for k_r taken from the literature together with calculated values of β^2 and α_{exp} . For two different

⁵ This is because, at the cone angle of 80° , the axial H atoms are nearly in van der Waals' contact with each other.

TABLE 1. Restricted free rotor model with correction for electron overlap

Radicals	$\log k_r$ (exp) (L/mol-s) ^a	$r^\ddagger$ (Å) (calcd)	$\log g_e Z_r$ (L/mol-s) ^b	α^c (exp)	$\beta_i \beta_j$ (calcd)	β_i (calcd)
2 CH ₃	10.4	5.37	10.75	0.34	0.30	0.55
2 C ₂ H ₅	10.0	5.29	10.60	0.20	0.18	0.42
2 i-C ₃ H ₇	9.7	5.24	10.50	0.13	0.10	0.31
2 <i>t</i> -Bu	9.1	5.19	10.45	0.028	0.029	0.17
Me - <i>t</i> -Bu ^d	10.2 ± 0.3	5.30	11.05	0.12	0.10	0.55/0.19
2 Allyl	9.9	4.72	10.75	0.12	0.16	0.4
2 CF ₃	9.7	5.44	10.55	0.15	0.14	0.38
2 CCl ₃	9.6	5.34	10.30	0.12	0.07	0.27
Cl + CCl ₃	9.8	6.02	10.65	0.23	0.27	0.27

^a Experimental values from ref. 3.^b Calculated using $r^\ddagger$ (calcd) and $g_e = 1/4$. $Z_r = \pi(r^\ddagger)^2 \bar{v} / \sigma$ where $\bar{v}$ = relative velocity at 300 K and $\sigma = 2$ for identical radicals.^c From experimental data and eq. [2].^d Using dissociation rate constant of ref. 10, of a bond dissociation energy of 83.5 kcal/mol, and an estimated $\Delta S_{300}^\ddagger = -45 \pm 1.5$ eu to calculate k_r at 800 K.

radicals recombining, $\dot{R}_i$ and $\dot{R}_j$, we use the product $\beta_i \beta_j$.

tert-Butyl radical recombination

Among the lowest values observed for the rate of radical recombination is that for *t*-Bu. While some experimental values have shown an alarming spread, the most recent and direct measurements are centered about k_r (*t*-Bu) = 1×10^9 L/mol-s with an uncertainty of about a factor of 2 (3). We calculate $r^\ddagger = 5.19$ Å using a value of V_0 estimated at about 77 kcal/mol. This gives $\langle Z_r \rangle = 3.6 \times 10^{10}$ L/mol-s and hence $\alpha = 0.028$ and $\beta = 0.16$. If we construct a bent model for *t*-Bu with a cone angle of 80°, about half-way between tetrahedral and a flat model, we estimate an active surface fraction β equal to 0.17. We do not include the back side of the *tert*-butyl but only the open side.⁵ We also fix the geometry of the H-atoms as corresponding to three axial and six equatorial. This yields $\beta^2 = 0.029$, in excellent agreement with the experimental result. Volumes for ethyl and *i*-propyl may be calculated in similar fashion or else by interpolating the values of β between *tert*-butyl and methyl. Table 1 shows values estimated in this fashion. The agreement with experiment, as evidenced by the comparison of β^2 or $\beta_i \beta_j$ with α , is within the experimental uncertainty for all of these systems.

The effect of temperature on k_r is complex. Overall, the model would appear to predict an increase in k_r varying as $T^{1/6}$. This arises from an increased collision velocity going as $T^{1/2}$ and a decreased collision cross section going as $b^2 \propto T^{-1/3}$. In the range from 300 K to 600 K, over which most measurements have been made, the increase would come to a factor of $2^{1/6} = 1.12$, too small an effect to be measured. However, as $b^\ddagger$ decreases, the collision radius decreases with it and the projected van der Waals' area of the bound atoms such as H in CH₃ becomes a larger part of the inactive surface. Hence β decreases and in the case of CH₃ radicals, which are simpler than most, we estimate a net decrease in recombination rate of the order of $T^{-1/6}$ or about 12% at 600 K.

For bulkier radicals such as *tert*-butyl the situation must be examined more closely. The collision radius we have calculated, of $5.19/2 \approx 2.60$ Å at 300 K, is smaller than the largest van der Waals' dimension, which is about 3.20 Å. Thus the methyl groups project somewhat beyond the surface of our effective collision sphere and the calculated value of β^2 in Table 1 ignores this. In those conformations where these methyl boundaries touch each other there is an added van der

Waals' constraint on the orientation for which a rough estimate reduces β^2 by about 8% at 300 K. At 600 K, with $b/2$ reduced by 12% to 2.32 Å, the van der Waals' repulsion is appreciably larger.

Some efforts have been made to measure CH₃ recombination in shock tubes near 1500 K. At this temperature $b = 4.11$ Å and $\beta^2 \leq 0.18$, a reduction of a factor of 3 in active area over that at 300 K. However, the effect on k_r is only a reduction of a factor of 2.3 because of the increase in collision rate. No firm data are available for comparison (8).

For many radicals, efforts have been made to deduce recombination rate constants from Arrhenius *A*-factors for the reverse reaction, measured at high temperatures, and the overall entropy change in the reaction. In the case of *tert*-butyl radicals, at 1000 K we deduce $b^\ddagger = 4.26$ Å and there is now considerable restriction in the transition state, both on the mutual orientation of the two radicals and on their internal rotation. In parallel conformation the methyl groups on the opposing *t*-Bu radicals are only 5.3 Å apart (assuming tetrahedral geometry), yielding an effective active area of about 0.07 of the surface. The recombination rate would be reduced by a factor of about 6 compared to its value at 300 K. Data in the range 700–800 K suggest that this may be indeed the case (6).

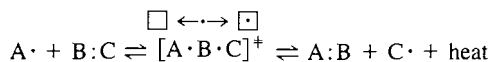
Radical disproportionation; activation energy

Radical disproportionation is generally a very exothermic reaction with no activation energy and very large rate constants. One which has been well studied, relative to recombination, is that between C₂H₅ radicals. The ratio of rates of disproportionation to recombination, k_d/k_r , is 0.14 over a very broad temperature range (3) and is also relatively insensitive to solvent or gas phase. The reaction is exothermic by 61 kcal/mol and if we adopt $k_r = 1 \times 10^{10}$ L/mol-s, then $k_d = 1.49 \times 10^9$ L/mol-s within the same accuracy with which k_r is known. Values for *i*-propyl and *tert*-butyl are somewhat higher, by factors of about 1.5 and 2 respectively (3). Since these are metathetical reactions, it is instructive to compare them with the known metathesis of radicals with molecules.

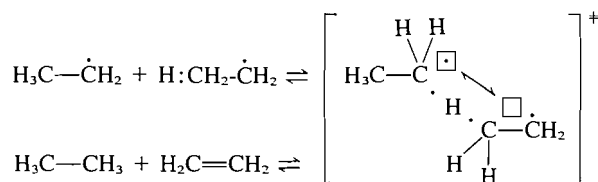
Most H-atom metathesis of alkyl radicals with hydrocarbons have intrinsic activation energies near 10 kcal, 90% of the known values lying within ± 3 kcal of this number. They are not very sensitive to the exothermicity of the reaction except when $\Delta H \sim 0$, such as for CH₃ + CH₄, when they tend to reach maximum values near 14 or 15 kcal (3). Because C—H bond

strengths lie in the range 84–105 kcal/mol, the exothermicities of metatheses are in the range 0–20 kcal/mol, considerably less than that for disproportionation.

It is of interest to try to understand the reason that radical disproportionations have zero activation energy. In the usual metathesis reaction, we change from the reactants in which we have two bonding electrons in the σ -bond being attacked to the tight transition state in which we have two one-electron bonds. Thus the number of bonding electrons is constant. The extra

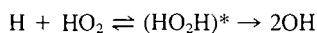


electron, initially on the attacking radical in the transition state is shared between the entering and leaving groups or atoms. It has been found that if either or both of these is a strong electrophile, the activation energy is thereby decreased (7). It has been suggested that the intrinsic activation energy in these reactions arises from the repulsion between the nonbonding electron and one of the two bonding electrons with which it has a similar spin orientation, essentially a type of triplet repulsion. The situation in disproportionation shows an interesting difference. Again, using two ethyl radicals, we can picture the electron distribution as:



The nonbonding electron can be displaced into an empty orbital associated with the near-neighbor carbon atom as shown, or even begin to participate in the formation of the π -bond, a bonding orbital. This near-neighbor carbon is thus acting as an electrophile and converting a nonbinding electron into a bonding electron. However one interprets such a model, it seems quite reasonable that the activation energy is lowered relative to metathesis and not surprising that it is zero, considering the large exothermicity of the overall reaction.

There are not too many cases reported of disproportionation between radicals involving transfer of atoms other than H. In the few cases that are known of Cl atom transfer, the activation energies reported are either zero or not more than 0.5 kcal (3). A number of reactions are classified as metatheses which are really radical recombinations followed by dissociation. Thus the reaction $\text{H} + \text{HO}_2 \rightarrow 2\text{OH}$, while overall involving O atom transfer, is really a radical recombination.



Radical disproportionation A-factors

The Arrhenius A-factor for disproportionation reactions turn out to be anomalously large compared to normal metathesis reactions. Thus, the tight transition state used to characterize metathesis yields A-factors of the order of 10^7 to 10^8 per H-atom, at least a factor of 10 lower than the values observed for disproportionation. If we calculate the A-factor for ethyl disproportionation by the usual procedure (Table 2), we estimate a value of $10^{7.9}$ L/mol-s, about a factor of 20 lower than the observed value.

Moreover, the usual tight transition state does not give us much flexibility. With 2 one-electron bonds in the $\text{C}\cdot\text{H}\cdot\text{C}$

configuration, the neighboring groups attached to the carbon atoms are at most about 2.8 Å apart (using $r(\text{C}\cdot\text{H}) = 1.4$ Å) and hence very close to van der Waals' contact. Bending the $\text{C}\cdot\text{H}\cdot\text{C}$ angle by 40° would achieve this if only H atoms were attached to the carbon atoms.

An even more dramatic view of these A-factors is afforded by the A-factor for the reverse reaction. In the case of ethyl disproportionation this would be the metathesis of C_2H_6 with C_2H_4 :



The entropy increase in this reaction is 9.4 eu so that, using $10^{9.1}$ L/mol-s as the A factor for the reverse reaction, we calculate $A = 10^{11.2}$ L/mol-s! If we assign the two partners a collision diameter of 5 Å, the A-factor would correspond to the total collision rate at 300 K!

Transition state theory is of course a more sophisticated picture of collision theory. The low A-factors calculated for metathesis merely reflect the steric inhibition placed on a possible collision. One can make a direct collision calculation which demonstrates this. If we require a 1.4 Å separation in the $\text{C}-\text{H}$ bond to be formed, then we can calculate the collision rate for such an event subject to steric constraints. For $2\text{C}_2\text{H}_5$, assuming 6 abstractable H atoms (3 on each ethyl), we can calculate $k_d = 3.8 \times 10^{10} \beta_H \beta_C$ L/mol-s (for singlet collision). Here $\beta_H \beta_C = \alpha$, the steric requirements on the possible orientation of each partner in the collision. If we allow each partner to undergo independent rotations about a fixed ($\text{C}-\text{H}$) collision geometry equal to an azimuthal deflection of 30° , we can calculate $\alpha = 4.5 \times 10^{-3}$ and $k_d = 1.8 \times 10^8$ L/mol-s. This is about 2.5 times larger than the transition state model because 30° is too high a flexibility for the assigned ($\text{C}-\text{H}$) contact distance. If we had restricted this to 24° we would have recovered the transition state result. It is still about a factor of 10 less than the observed k_d (C_2H_5).

All of these results imply that the interaction between two ethyl radicals must occur at larger separations than we have been calculating. We gain some idea by looking at the product ($\sigma^2 = \beta_H \beta_C$). The chosen values gave us a product 10 times too small. Hence $\sigma = \beta_i$ ($i = 1, 2$) must be increased about three-fold to reproduce the data. For example, if we make $r(\text{C}-\text{H}) = \sigma = 4.3$ Å, keeping β_i fixed or increasing both appropriately, we could reproduce k_d . At distances like 4.3 Å, however, van der Waals' interactions between hydrocarbons are negligible compared to RT and there is no reason why this distance should be significant. Let us suppose, however, that free radicals have a net dipole moment. If they are not exactly planar we would expect a net dipole along the electron orbital. Since the $\text{C}-\text{H}$ bond has a small dipole moment we would then expect a dipole-dipole interaction which could be represented by:

$$[12] \quad E_{d-d} = \frac{2\mu_1\mu_2\cos\gamma}{r_{12}^3}$$

where γ is the azimuth angle between the $\text{C}-\text{H}$ bond and the electron orbital dipole. Reasonable dipole moments for these would be 0.32 D for the $\text{C}-\text{H}$ bond and 1.5 D for the $\text{C}\cdot$ moment. Let us assume that these are centered in the midpoint of the $\text{C}-\text{H}$ bond and the carbon nucleus, respectively, and both oriented along their respective bond directions. Such an interaction is now relatively long range and begins to resemble the recombination model. We can look for the distance at which this interaction is of the order of RT at 300 K, assuming

TABLE 2. Tight transition state calculation of A-factor for ethyl disproportionation
 $2\text{C}_2\text{H}_5 \rightleftharpoons [\text{CH}_3\text{---CH}_2\cdot\text{H}\cdot\text{CH}_2\text{---CH}_2]^\ddagger$

Degree of freedom	Correction to $\Delta S^\ddagger_{300}$ (eu) ^a
Symmetry; $R \ln 3$	2.2
External rotation, $R \ln 2$	1.4
1 barrier $V_0 \rightarrow 0$ kcal	1.4
1 barrier $V_0 \rightarrow 0.5$ kcal	1.3
1 barrier $V_0 \rightarrow 1.5$ kcal	0.8
4 C—C—C bond decrease by 0.7 in frequency	2.8
Reaction coordinate: (C—C) stretch at 900 cm^{-1}	-0.2
	Total 9.7 eu
	$\Delta S^\ddagger = -33.0$ eu (correct to concentration units by +8.4 eu)

$$A_{300} = \left(\frac{ekT_0}{h}\right)e^{\Delta S^\ddagger/R} = 10^{7.9} \text{ L/mol-s}$$

^a $\Delta S^\ddagger$ for model reaction using $n\text{-C}_4\text{H}_{10}$ to represent transition state: $\Delta S^\ddagger_{300} = -42.7$ eu

TABLE 3. Steric factors for radical disproportionation

Radical pair	$\log k_d^a$ (L/mol-s)	Reduced mass (amu)	$(\beta_C/\beta_{Me})_{\text{obs}}$	$(\beta_R/\beta_{Me})_{\text{calcd}}^b$
$\text{CH}_3 + \text{C}_2\text{H}_5$	9.15	10.0	1	1.00 (0.55)
$\text{C}_2\text{H}_5 + \text{C}_2\text{H}_5$	9.15	14.5	0.8	0.76 (0.46)
$i\text{-Pr} \times i\text{-Pr}$	9.4	21.5	0.6	0.56 (0.31)
$t\text{-Bu} \times t\text{-Bu}$	9.5	28.5	0.42	0.34 (0.19)

^a Data from ref. 3 except for $\text{CH}_3 + \text{C}_2\text{H}_5$ for which $k_d/k_r = 0.04$ is known and we have estimated k_r from the geometric mean law $k_{r12} = 2(k_{r11}k_{r22})^{1/2}$.

^b Steric factors taken from Table 1 for radical recombination. Absolute fractions for each radical are shown in parentheses (from Table 1).

that $\gamma \leq 30^\circ$. We find (eq. [3]) that $r_{12} \leq 2.75 \text{ \AA}$ for a $\text{C}\cdots\text{H}$ distance of $2.20 \text{ \AA}$ which corresponds to a $\text{C}\cdots\text{H}\cdots\text{C}$ distance of about $3.9 \text{ \AA}$. At these larger distances there is much less steric inhibition and an azimuthal angle between the two ethyls of even up to 40° can be accommodated. In fact $r^\ddagger(\text{C}\cdots\text{H}) = 2.2 \text{ \AA}$, together with azimuthal angles of 40° , will reproduce k_d for ethyls.

In the absence of precise information on the $\dot{\text{C}}$ moment let us proceed empirically. In Table 3 are collected some experimental data on alkyl radical disproportionations. We can represent k_d in the same way in which we treated k_r :

$$[13] \quad k_d = \frac{1}{4}Z_d\alpha = \frac{1}{4}Z_d\beta_C\beta_H$$

Z_d is a "hard sphere" collision rate $= \pi\sigma_d^2 \bar{v} = \pi\sigma_d^2 (8RT/\pi\mu)^{1/2}$ in which μ is the reduced mass of the collision and σ_d the collision diameter. For nonidentical radicals two possible reactions are possible. Z_d includes the correction for the fact that disproportionation of two identical radicals can occur from either one through an asymmetric transition state. β_C and β_H are the fractions of solid angle available to the collision pair at the carbon radical (β_C) and abstracted H respectively. Without detailed knowledge, there are three parameters in eq. [13], σ_d , β_C , and β_H . We can reduce these by making a relative comparison.

Let us assume that in the cases listed in Table 3, the same number of H atoms in each CH_3 group are available for disproportionation and that β_H is a transferable constant for a methyl group. Ethyl, *i*-Pr, and *tert*-butyl then have β_H factors

in the simple ratio 1:2:3 reflecting the number of available methyl groups. If we further assume that the dynamical, long range interaction is the same, then σ_d is the same for all pairs. By taking ratios of k_r for the different pairs relative to methyl, correcting for reduced masses and numbers of methyls, we can deduce ratios of β_R/β_{Me} which are shown in the 4th column of Table 3. In the last column of Table 3 are shown the ratios of fractions of surface, relative to CH_3 , which are active in recombination. The two columns agree to within the experimental uncertainty.

This agreement suggests that the long range forces active in disproportionation are similar to the exchange forces active in recombination. This would help explain the fact that the ratios k_d/k_r show little temperature variation. The developing C—H bond would resemble a hydrogen bond with exchange character. A simple ad hoc model may then be constructed to reproduce the value of k_d .

Let us assume that at $r(\text{C}\cdots\text{H}\cdots\text{C}) = 4.0 \text{ \AA}$, the critical interaction is of the order of RT and that the same fraction of active area of the carbon center that we computed for recombination is also active in disproportionation. Then we can compute from the data in Table 3 that, for the methyl group, $\beta_H = 0.067$. This corresponds to a conical section of the methyl sphere with an azimuth angle of 30° , small enough to suggest that only one of the three H atoms is active in disproportionation. If all three were active, then for each the azimuth angle would be reduced to 17° . From a chemical point of view, it would be reasonable to expect attack on axial H-atoms which are in the *trans* position to the odd electron. This would facilitate π -bond formation.

1. E. GORIN. *J. Chem. Phys.* **7**, 263 (1979); **7**, 642 (1979).
2. S. W. BENSON. *Thermochemical kinetics*. 2nd ed. John Wiley and Sons, New York. 1976.
3. J. A. KERR and S. J. MOSS. *CRC handbook of bimolecular and termolecular gas reactions*. Vol. II. CRC Press, Inc., Boca Raton, FL. 1981.
4. M. H. BAGHAL-VAYJOEE, A. COLUSSI, and S. W. BENSON. *J. Am. Chem. Soc.* **100**, 3214 (1978); *Int. J. Chem. Kinet.* **11**, 147 (1979).
5. D. L. BAULCH and J. DUXBURY. *Combust. Flame*, **37**, 313 (1980).
6. K. Y. CHOO, P. L. BEADLE, L. W. PISKIEWICZ, and D. M. GOLDEN. *Int. J. Chem. Kinet.* **8**, 45 (1976).
7. Z. ALFASSI and S. W. BENSON. *Int. J. Chem. Kinet.* **5**, 879 (1973).
8. K. GLÄNZER, M. QUACK, and J. TROE. 16th Symp. Int. Combust. Proc. 949 (1976); J. TROE. *Am. Rev. Phys. Chem.* **29**, 223 (1978).
9. S. W. BENSON. *The foundations of chemical kinetics*. Reprint Ed. Krieger Publ. Co., Malabar, FL. 1982. p. 144.
10. J. ENGLE, A. COMBE, M. LETORT, and M. NICLAUSE. *C.R. Acad. Sci.* **244**, 453 (1957).